

TRIG IDENTITIES

$$\begin{aligned} & \star (1 - 2\sin^2(\theta)) \\ & = (1 + \sqrt{2}\sin(\theta))(1 - \sqrt{2}\sin(\theta)) \end{aligned}$$

RECIPROCAL

$$\begin{aligned} \csc(x) &= \frac{1}{\sin(x)} \\ \sec(x) &= \frac{1}{\cos(x)} \\ \cot(x) &= \frac{1}{\tan(x)} \end{aligned}$$

HALF-ANGLE

$$\begin{aligned} \sin\left(\frac{x}{2}\right) &= \pm \sqrt{\frac{1 - \cos(x)}{2}} \\ \cos\left(\frac{x}{2}\right) &= \pm \sqrt{\frac{1 + \cos(x)}{2}} \\ \tan\left(\frac{x}{2}\right) &= \pm \sqrt{\frac{1 - \cos(x)}{1 + \cos(x)}} \\ &= \frac{\sin(x)}{1 + \cos(x)} = \frac{1 - \cos(x)}{\sin(x)} \end{aligned}$$

SUM-TO-PROD

$$\begin{aligned} \sin(x) + \sin(y) &= 2\sin\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right) \\ \sin(x) - \sin(y) &= 2\sin\left(\frac{x-y}{2}\right)\cos\left(\frac{x+y}{2}\right) \\ \cos(x) - \cos(y) &= -2\sin\left(\frac{x+y}{2}\right)\sin\left(\frac{x-y}{2}\right) \\ \cos(x) + \cos(y) &= 2\cos\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right) \end{aligned}$$

QUOTIENT

$$\begin{aligned} \tan(x) &= \frac{\sin(x)}{\cos(x)} \\ \cot(x) &= \frac{\cos(x)}{\sin(x)} \end{aligned}$$

PYTHAGOREAN

$$\begin{aligned} \sin^2(x) + \cos^2(x) &= 1 \\ 1 + \tan^2(x) &= \sec^2(x) \\ 1 + \cot^2(x) &= \csc^2(x) \end{aligned}$$

EVEN/ODD

$$\begin{aligned} \sin(-x) &= -\sin(x) \\ \cos(-x) &= \cos(x) \\ \tan(-x) &= -\tan(x) \\ \cot(-x) &= -\cot(x) \\ \sec(-x) &= \sec(x) \\ \csc(-x) &= -\csc(x) \end{aligned}$$

CO-FUNCTION

$$\begin{aligned} \sin\left(\frac{\pi}{2} - x\right) &= \cos(x) \\ \cos\left(\frac{\pi}{2} - x\right) &= \sin(x) \\ \tan\left(\frac{\pi}{2} - x\right) &= \cot(x) \\ \cot\left(\frac{\pi}{2} - x\right) &= \tan(x) \\ \sec\left(\frac{\pi}{2} - x\right) &= \csc(x) \\ \csc\left(\frac{\pi}{2} - x\right) &= \sec(x) \end{aligned}$$

DOUBLE ANGLE

$$\begin{aligned} \sin(2x) &= 2\sin(x)\cos(x) \\ &= \frac{2\tan(x)}{1 + \tan^2(x)} \\ \tan(2x) &= \frac{2\tan(x)}{1 - \tan^2(x)} \\ \cos(2x) &= \cos^2(x) - \sin^2(x) \\ &= 2\cos^2(x) - 1 \\ &= 1 - \sin^2(x) = \frac{1 - \tan^2(x)}{1 + \tan^2(x)} \end{aligned}$$

TRIPLE ANGLE

$$\begin{aligned} \sin(3x) &= 3\sin(x) - 4\sin^3(x) \\ \cos(3x) &= 4\cos^3(x) - 3\cos(x) \\ \tan(3x) &= \frac{3\tan(x) - \tan^3(x)}{1 - 3\tan^2(x)} \\ \csc(3x) &= \frac{3\sin(x) - 4\sin^3(x)}{1 - 3\tan^2(x)} \\ \sec(3x) &= \frac{4\cos^3(x) - 3\cos(x)}{1 - 3\tan^2(x)} \\ \cot(3x) &= \frac{3\tan(x) - \tan^3(x)}{1 - 3\tan^2(x)} \end{aligned}$$

PROD-TO-SUM

$$\begin{aligned} \cos(x)\cos(y) &= \frac{1}{2}(\cos(x-y) + \cos(x+y)) \\ \sin(x)\cos(y) &= \frac{1}{2}(\sin(x+y) + \sin(x-y)) \\ \sin(x)\sin(y) &= \frac{1}{2}(\cos(x-y) - \cos(x+y)) \\ \cos(x)\sin(y) &= \frac{1}{2}(\sin(x+y) - \sin(x-y)) \end{aligned}$$

